

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
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Relevant Experience

HP has a background in both engineering and science; 28 yrs as a snow scientist, and 20 yrs developing FMCW radar systems. He has successfully led 27 research grants from federal and state agencies. He has supported SWE remote sensing efforts in Europe (ESA CoreH2O) as well as the U.S. (CLPX I/II/III), and was the Project Scientist for the NASA SnowEx Mission from 2018-2021, and on the leadership team from 2016-2023.

Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington, Thesis: *Wet snow densification*

Selected Professional Experience

- 08/2018-8/2021 Project Scientist, NASA SnowEx Mission
- 08/2016-07/2018 Co-lead of Field Campaign, NASA SnowEx 2017 Mission
- 08/2022-present Professor, Department of Geosciences, Boise State University

Selected Honors and Awards

- 2020 BSU Faculty Excellence Award
- 2019 NASA Instrument Incubator Program Award, NASA, 1st in Idaho history
- 2019 2018 WRR Editors' Choice Award: Hedrick et al., 2018
- 2018 NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
- 2012 Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
- 2010-13 New Investigator Program Award, NASA, 1st in Idaho history
- 2010 Young Scientist Award, Cryosphere Section, American Geophysical Union
- 2009 Top Reviewer Award, Cold Regions Science and Technology (CRST)
- 2008 U.S. Young Scientist Award, International Union of Radio Science (URSI)

Selected Relevant Peer-Reviewed Publications

[total=105, h-index=36, i10-index=80, citations=4450]

- 2025 Alabi, I., **Marshall, H.P.**, Mead, J., and Trujillo, E. (2025). Advancing terrestrial snow depth monitoring with machine learning and l-band insar data: a case study using nasa's snowex 2017 data. *Frontiers in Remote Sensing*, 5(1481848):doi: 10.3389/frsen.2024.1481848
- Brangers, I., De Lannoy, G., **Marshall, H.P.**, Dunmire, D., Bonnell, R., Cox, B., Cappelle, J., Brad Baxter, W., and Lievens, H. (2025). C-band radar measurements in a snow-covered boreal forest environment. *IEEE Geoscience and Remote Sensing Letters*, 22:1–5

- Tarricone, J., Palomaki, R., Rittger, K., Nolin, A., **Marshall, H.P.**, and Vuyovich, C. (2025). Investigating the impact of optical snow cover data on l-band insar snow water equivalent retrievals. *Journal of Remote Sensing*, 5:1–18, DOI: 10.34133/remotesensing.0682
- 2024 Oveisgharan, S., Zinke, R., Hoppinen, Z., and **Marshall, H.P.** (2024). Snow Water Equivalent Retrieval Over Idaho, Part A: Using Sentinel-1 Repeat-Pass Interferometry. *The Cryosphere*, 18:559–574, doi.org/10.5194/tc-18-559-2024
- Hoppinen, Z. M., Oveisgharan, S., **Marshall, H.P.**, Mower, R., Elder, K., and Vuyovich, C. (2024). Snow Water Equivalent Retrieval Over Idaho, Part B: Using L-band UAVSAR Repeat-Pass Interferometry. *The Cryosphere*, 18:575–592, doi.org/10.5194/tc-18-575-2024
- 2023 Bonnell, R., McGrath, D., Hedrick, A. R., Trujillo, E., Meehan, T. G., Williams, K., **Marshall, H.P.**, Sexstone, G., Fulton, J., Ronayne, M. J., Fassnacht, S. R., Webb, R. W., and Hale, K. E. (2023). Snowpack relative permittivity and density derived from near-coincident lidar and ground-penetrating radar. *Hydrological Processes*, 37(10):e14996
- Tarricone, J., Webb, R., **Marshall, H.P.**, Nolin, A., and Meyer, F. (2023). Estimating snow accumulation and ablation with L-band InSAR. *The Cryosphere*, 17(5):1997–2019, doi.org/10.5194/tc-17-1997-2023
- 2022 Lievens, H., Brangers, I., **Marshall, H.P.**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc-16-159-2022
- Lund, J., Forster, R., Deeb, E., Liston, G., Skiles, M., and **Marshall, H.P.** (2022). Interpreting Sentinel-1 SAR backscatter signals of snowpack surface melt/freeze, warming, and ripening, through field measurements and physically-based snowmodel. *Remote Sensing*, 14:4002, https://doi.org/10.3390/rs14164002
- McGrath, D., Bonnell, R., Zeller, L., Olsen-Mikitowicz, A., Bump, E., Webb, R., and **Marshall, H.P.** (2022). A time-series of snow density and snow water equivalent observations derived from the integration of GPR and UAV SfM observations. *Frontiers in Remote Sensing*, 3:886747, doi:10.3389/frsen.2022.886747
- Tsang, L., Durand, M., Derksen, C., Barros, A. P., Kang, D.-H., Lievens, H., **Marshall, H.P.**, Zhu, J., Johnson, J., King, J., Lemmetyinen, J., Sandells, M., Rutter, N., Siqueira, P., Nolin, A., Osmanoglu, B., Vuyovich, C., Kim, E. J., Taylor, D., Merkouriadi, I., Brucker, L., Navari, M., Dumont, M., Kelly, R., Kim, R. S., Liao, T.-H., and Xu, X. (2022). Review article: Global monitoring of snow water equivalent using high frequency radar remote sensing. *The Cryosphere*, 16(9):3531–3573
- 2021 **Marshall, H.P.**, Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852
- 2019 Lievens, H., Demuzere, M., **Marshall, H.P.**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Giroto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12

- McGrath, D., Webb, R., Shean, D., Bonnell, R., **Marshall, H.P.**, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during NASA's 2017 SnowEx Campaign: Comparison to in situ, airborne, and satellite observations. *Water Resources Research*, 55(11):10026–10036, doi:10.1029/2019WR024907
- 2008 **Marshall, H.P.** and Koh, G. (2008). FMCW radars for snow research. *Cold Regions Science and Technology*, 52(2):118–131
- 2007 **Marshall, H.P.**, Schneebeli, M., and Koh, G. (2007). Snow stratigraphy measurements with high-frequency FMCW radar: Comparison with snow micro-penetrometer. *Cold Regions Science and Technology*, 47(1-2):108–117